

Exercise Sheet 1

Deadline for submission: Wednesday, 1 October 2025 at 08:00.

Discussed in exercise session: Friday, 3 October 2025 at 10:15.

Exercise 1.1 - Solution(s) of a Linear System of Equations

Consider the following linear system of equations in two-dimensional space

$$\begin{cases} 2x_1 - 2x_2 = -2 \\ 4x_1 - x_2 = 2. \end{cases} \quad (1)$$

- a) Rewrite the linear system of equations (1) in matrix-vector notation, that means in the form of $A\mathbf{x} = \mathbf{b}$. Define additionally the matrix A as well as the vectors $\mathbf{x}$ and $\mathbf{b}$.
- b) Define the column space $C(A)$ for the matrix A . Is the equation (1) solvable? Explain.
- c) Check if the vectors $(2, 4)^T$ and $(-2, -1)^T$ of the column space $C(A)$ are linearly independent and define a basis $\mathcal{B}$ (of the column space).
- d) Determine the solution of equation (1) by using the Gaussian elimination method with back substitution or the Gauss-Jordan elimination, and display it graphically once in row and once in column view. How many solutions can you find?
- e) Change one equation in the linear system of equations (1) such that (i) the new linear system of equations has no solutions, and (ii) such that the new linear system of equations has infinitely many solutions. Display both graphically in row view.

Exercise 1.2 - Gauss Elimination and Gauss Jordan Algorithm

We would like to solve the following linear system of equations

$$\begin{cases} x_2 - x_3 = 1 \\ x_1 + 2x_2 - 3x_3 = 2 \\ -2x_1 + x_2 = -3. \end{cases} \quad (2)$$

- a) Rewrite the linear system of equations (2) in matrix-vector notation, that means in the form of $A\mathbf{x} = \mathbf{b}$, and define the augmented matrix $\mathbf{A}_{\text{aug}}$.
- b) In lecture we saw three elementary matrices R_{ij} , $R_i(m)$ and $R_{ij}(m)$ which we can apply to our augmented matrix $\mathbf{A}_{\text{aug}} = [\mathbf{A}, \mathbf{b}, \mathbf{I}_3]$ without changing the solution. In the first step, we are interested in the solution by applying the Gaussian elimination algorithm with back substitution. Apply these three forms of elementary matrices by left-multiplication to the matrix $\mathbf{A}_{\text{aug}}$, and solve the system by back substitution.
- c) In the second step, we want to apply the Gauss-Jordan elimination algorithm. Therefore, we need to proceed with the elementary operations by left-hand side multiplication to our augmented matrix $\mathbf{A}_{\text{aug}}$ until we get an identity matrix at the left-hand side, this means we get a matrix which has the form of $[\mathbf{I}_3, \mathbf{x}, \mathbf{A}^{-1}]$. Determine the solution again after the Gauss-Jordan elimination. (*Solution vector should be the same as before*)